

Development of a Bayesian Optimization algorithm to explore magnetic field topology for radiation dose reduction in crewed space missions shielding.

The problem of radiation shielding is of paramount importance in crewed space missions: the limitations of damages to the astronauts' healths puts limits on the duration of a crewed mission. At this time, NASA duration for ISS missions is 6 months [1]. This cap makes unfeasible at the current stage the possibility of planning crewed mission to explore other solar system planets.

There are two types of radiation shielding: passive and active. Passive shielding operates by placing some shielding material that absorbs radiation by particle interaction. This is the type of shielding that the most research has been conducted about.

Active radiation shielding relies on a magnetic field to deflect incoming particles via Lorentz force. Some designs have been proposed, like in [2–4]. Also, preliminary studies have been conducted by NASA concluding that using magnetic fields generated by high temperature superconducting coils is possible, while the proposed designs do not offer significant improvements over passive shielding [5]. At the current state, no algorithms for BO in magnetic field optimization have been proposed in the reviewed literature.

In the current literature, the most widely used method for magnetic field optimization is topology optimization, with applications to finding the best coil distribution for the largest uniform field in [6] and in reverse magnetostatics to find optimal configuration of coils to generate a target field in [7]. This may prove useful to reverse engineer an optimal solution from Bayesian Optimization. Given the experience of the RMIT Space Physics Group in this challenge, as in [8–10], this research proposes using a statistical optimization model to explore exotic field topologies for minimizing the received radiation dose in Monte Carlo simulations. Due to the complex and non-linear nature of dependencies between field topology and shielding capabilities, no magnetic field optimization has been applied to this task. However we feel that the huge exploration capabilities, even when searches are expensive, and the ability to operate in a black box scenario of BO may provide an important insight in designing the shield configuration.

Research Questions:

1. Is there an optimal field topology that minimizes the probability of particle penetration and can it be found via Bayesian Optimization?
2. What low-dimensional parametrizations of magnetic fields produces a physically meaningful optimum while enabling efficient Bayesian Optimization?
3. What theoretical guarantees can be established for Gaussian Process-based Bayesian Optimization in this PDE-constrained electromagnetic design problem, where the objective is noisy, non-smooth, and induced by stochastic charged-particle transport?

Methods

The main mathematical tool is the Gaussian Process-based Bayesian Optimization, which tries to minimize the functional $\mathcal{J}(\theta)$ by observing data from it and without having to know [11]. The algorithm works as follow:

Algorithm 1 Basic BO minimization

Place a Gaussian process prior on $\mathcal{J}$.
Observe $\mathcal{J}$ at an n_0 number of points and set $n = n_0$.
while ($n \leq N$) **do**
 Update posterior probability distribution of $\mathcal{J}$, using all available data.
 Let θ_n be a maximizer of an acquisition function over θ .
 Observe $y_n = \mathcal{J}(\theta_n)$.
 Increment n .
end while
Return point with highest evaluated $\mathcal{J}(\theta)$

The first task will be to find a parametrization of the $\mathbf{B}$ field that is both mathematically tractable and physically informative, given that BO works better in low dimension (max 50 params. approx.).

Then the algorithm will be implemented in a python code that will pass the field parameters to a C++ Monte Carlo simulations (the $\mathcal{J}$ functional), that will return the received radiation dose. The RMIT Space Physics group has provided the computational resources.

The RMIT Space Physics group has already developed simulators for this task that run on the particle physics software GEANT4, that can be used for the task.

A theoretical study on convergence guarantees and robustness under noise will be conducted in parallel, with the help of italian supervisors [12].

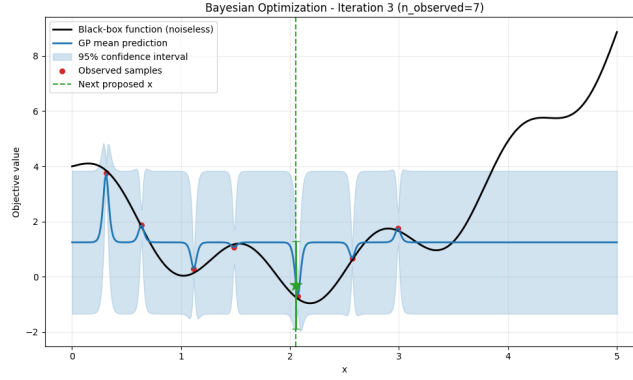


Figure 1: A mock-up BO at iteration 3.

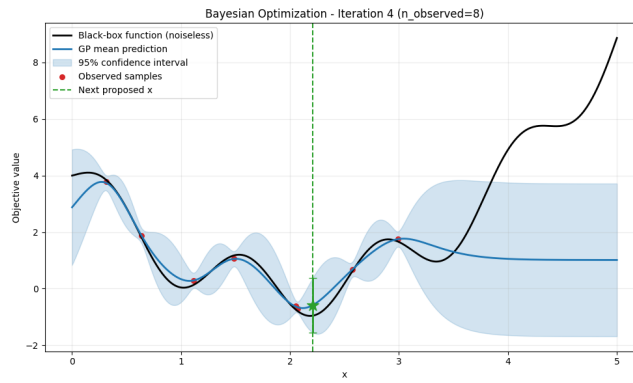


Figure 2: A mock-up BO at iteration 4.

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